

Generation Interconnection Request System Impact Study Report

For Queue Project Q12-0138

Maple Ave 100 kV
October 27, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0138. The Interconnection Customer proposes a 84.2 MW wind facility interconnecting at the Maple Ave 100 kV substation in the Illinois area. The request was received on June 16, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 103 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$147,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Beaver to Onancock Diesel Generation Facility 230 KV	Beaver to Eufaula 500 KV	101.3	605	11.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Beaver to Onancock Diesel Generation Facility 230 KV	Reconductor line	\$6,835,000	2% (\$147,000)
	Total allocated to project		\$147,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.